

LİNEER CEBİR 2019 GÜZ FINAL SORULARI

1-) A bir tekil olmayan bir matris. $3A = A^2 + AB$

$$B = \begin{bmatrix} 2 & 3 & -3 \\ -3 & 8 & -3 \\ -6 & 6 & -1 \end{bmatrix}$$

a) $A = ?$

b) A 'nin öz değerleri?

c) A 'nin öz vektörleri?

d) A köşegenleştirilip köşegenleştirilmediğini biliniz.

e) $A^{-1} = ?$ (Cayley-Hamilton teoreminden)

2-) $x_1 + x_2 + x_3 + x_4 + x_5 = 0$

$$2x_1 + x_2 - x_3 + x_4 = 0$$

a) Çözüm uzayının basını bulunuz.

b) " " ortogonal basını bulunuz.

c) " " orthonormal " "

3-) $L\left(\begin{bmatrix} 3 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $L\left(\begin{bmatrix} 4 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ -9 \\ 1 \end{bmatrix}$

a) L 'nin standart baza göre görüntüsünü bulunuz.

b) $\text{Ker } L = ?$

c) $\dim L(\mathbb{R}^n) = ?$

d) $\langle A, B \rangle = a_1b_1 + 2a_2b_2 + 3a_3b_3 + 4a_4b_4$ iç çarpım uzayı olup olmadığını göster.

a) A tekil olmayan bir matris ise $|A| \neq 0 \Rightarrow A^{-1}$ vardır

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$3A = A^2 + AB$ eşitliğini soldan A^{-1} ile çarparsak

$$\underbrace{3A^{-1}A}_{3I_3} = \underbrace{A^{-1}A^2}_{I_3 \cdot A} + \underbrace{A^{-1}AB}_{I_3 B} \Rightarrow 3I = A + B \Rightarrow A = 3I_3 - B$$

$$A = 3I_3 - B = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} - \begin{bmatrix} 2 & 3 & -3 \\ -3 & 8 & -3 \\ -6 & 8 & -1 \end{bmatrix} = \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix}$$

b) $|A - \lambda I| = 0$

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & -3 & 3 \\ 3 & -5-\lambda & 3 \\ 6 & -6 & 4-\lambda \end{vmatrix} = (1-\lambda) \begin{vmatrix} -5-\lambda & 3 \\ -6 & 4-\lambda \end{vmatrix} + 3 \begin{vmatrix} 3 & 3 \\ 6 & 4-\lambda \end{vmatrix} + 3 \begin{vmatrix} 3 & -5-\lambda \\ 6 & -6 \end{vmatrix}$$

$$= (1-\lambda) \cdot (-20 - 4\lambda + 5\lambda + \lambda^2 + 18) + 3(12 - 3\lambda - 18) + 3(-18 + 30 + 6\lambda)$$

$$= (1-\lambda) \cdot (\lambda^2 + \lambda - 2) + 3(-3\lambda - 6) + 3(6\lambda + 12)$$

$$= (1-\lambda) \cdot (\lambda-1) \cdot (\lambda+2) + 9(\lambda+2) = (\lambda+2) [9 - (\lambda-1)^2]$$

$$= (\lambda+2) \cdot (3 - (\lambda-1)) \cdot (3 + (\lambda-1))$$

$$= (\lambda+2) \cdot (4-\lambda) \cdot (2+\lambda) = (\lambda+2)^2 \cdot (4-\lambda)$$

$$\Rightarrow \lambda_1 = -2 \quad \lambda_2 = -2 \quad \lambda_3 = 4$$

c) $\lambda_1 = -2$ için $(A + 2I)X_1 = 0$

$$\begin{bmatrix} 3 & -3 & 3 \\ 3 & -3 & 3 \\ 6 & -6 & 4 \end{bmatrix} \begin{bmatrix} x_{11} \\ x_{12} \\ x_{13} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 3 & -3 & 3 & | & 0 \\ 3 & -3 & 3 & | & 0 \\ 6 & -6 & 4 & | & 0 \end{bmatrix} \xrightarrow{\substack{r_1(-1) \\ r_2(-1) \\ r_3(1/3)}} \begin{bmatrix} 0 & 0 & 0 & | & 0 \\ 1 & -1 & 1 & | & 0 \\ 0 & 0 & -2 & | & 0 \end{bmatrix} \xrightarrow{\substack{r_3(-1/2) \\ r_2(-1)}} \begin{bmatrix} 0 & 0 & 0 & | & 0 \\ 1 & -1 & 0 & | & 0 \\ 0 & 0 & 1 & | & 0 \end{bmatrix}$$

$$x_{11} = t \Rightarrow x_{12} = t \quad x_{13} = 0$$

$$X_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} t$$

$$\lambda_2 = -2 \text{ için de } X_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} t$$

$r = 2 < n = 3$
 $3 - 2 = 1$ def. bağıl

$$n_3 = 4 \text{ için } (A - 4I)x_3 = 0$$

$$\begin{bmatrix} -3 & -3 & 3 \\ 3 & -9 & 3 \\ 6 & -6 & 0 \end{bmatrix} \begin{bmatrix} x_{13} \\ x_{23} \\ x_{33} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} -3 & -3 & 3 & | & 0 \\ 3 & -9 & 3 & | & 0 \\ 6 & -6 & 0 & | & 0 \end{bmatrix} \xrightarrow{\substack{r_{21} \\ r_{31}}} \begin{bmatrix} -1 & -1 & 1 & | & 0 \\ 1 & -3 & 1 & | & 0 \\ 1 & -1 & 0 & | & 0 \end{bmatrix} \xrightarrow{\substack{r_{23}(-2) \\ r_{3}(-)}} \begin{bmatrix} -1 & -1 & 1 & | & 0 \\ 0 & -4 & 2 & | & 0 \\ 0 & -2 & 1 & | & 0 \end{bmatrix}$$

$$\xrightarrow{r_{13}} \begin{bmatrix} -1 & -1 & 1 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & -2 & -1 & | & 0 \end{bmatrix} \xrightarrow{r_{13}} \begin{bmatrix} -1 & 1 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & 2 & -1 & | & 0 \end{bmatrix} \rightarrow \begin{matrix} \text{rank} = 2 \\ n = 3 \end{matrix}$$

$3 - 2 = 1$ deg. bekl. çözüm

$$x_{32} = t \Rightarrow x_{31} = t \quad x_{33} = 2t$$

$$x_3 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} t$$

Öz vektörler ($t=1$ için)

$$x_1 = x_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

d) 3 farklı öz değer için 2 farklı öz vektör olduğu için köşegenleştirilemez. Ancak ügüleştirilebilir.

$$e) p(\lambda) = (\lambda + 2)^2 (4 - \lambda) = (\lambda^2 + 4\lambda + 4)(4 - \lambda) = 4\lambda^2 + 16\lambda + 16 - \lambda^3 - 4\lambda^2 - 4\lambda = 0$$

$$p(\lambda) = -\lambda^3 + 12\lambda + 16 = 0$$

Cayley-Hamilton dan;

$$-A^3 + 12A + 16I_3 = 0 \text{ olur. Esittiyi her iki tarafını } A^{-1} \text{ ile çarparsak,}$$

$$-A^2 + 12I_3 + 16A^{-1} = 0 \Rightarrow A^{-1} = \frac{A^2 - 12I_3}{16}$$

$$A^2 = \begin{bmatrix} 1 & -3 & 3 \\ 3 & -9 & 3 \\ 6 & -6 & 4 \end{bmatrix} \begin{bmatrix} 1 & -3 & 3 \\ 3 & -9 & 3 \\ 6 & -6 & 4 \end{bmatrix} = \begin{bmatrix} 1-9+18 & -3+19-18 & 3-9+12 \\ 3-19+18 & -9+25-18 & 9-19+12 \\ 6-18+24 & -18+10-24 & 18-18+16 \end{bmatrix} = \begin{bmatrix} 10 & -6 & 6 \\ 6 & 2 & 6 \\ 12 & -12 & 16 \end{bmatrix}$$

$$A^{-1} = \frac{\begin{bmatrix} 10 & -6 & 6 \\ 6 & 2 & 6 \\ 12 & -12 & 16 \end{bmatrix} - \begin{bmatrix} 12 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 12 \end{bmatrix}}{16} = \frac{\begin{bmatrix} -2 & -6 & 6 \\ 6 & -10 & 6 \\ 12 & -12 & 4 \end{bmatrix}}{16} = \begin{bmatrix} -1/8 & -3/8 & 3/8 \\ 3/8 & -5/8 & 3/8 \\ 3/4 & -3/4 & 1/4 \end{bmatrix}$$

$$x_1 + x_2 + x_3 + x_4 + x_5 = 0$$

$$2x_1 + x_2 - x_3 + x_4 = 0$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & | & 0 \\ 2 & 1 & -1 & 1 & 0 & | & 0 \end{bmatrix} \sim \begin{bmatrix} \underbrace{-1}_{r} & \underbrace{0}_{s} & \underbrace{2}_{t} & 0 & 1 & | & 0 \\ \underbrace{2}_{r} & \underbrace{1}_{s} & \underbrace{-1}_{t} & 1 & 0 & | & 0 \end{bmatrix} \quad \textcircled{2}$$

$$\text{rank } 2 \\ n = 5 \\ 5 - 2 = 3$$

a)

$$\begin{aligned} x_1 &= r \\ x_2 &= s \\ x_3 &= t \\ x_4 &= -2r - s + t \\ x_5 &= r + 2t \end{aligned}$$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -2 \\ 1 \end{bmatrix} r + \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \\ 0 \end{bmatrix} s + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \\ -2 \end{bmatrix} t$$

$$\text{Baz} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \\ -2 \end{bmatrix} \right\}$$

$\downarrow$ u_1 $\downarrow$ u_2 $\downarrow$ u_3

b)

$$v_1 = u_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

$$\langle v_1, v_1 \rangle = 1 + 0 + 0 + 4 + 1 = 6$$

$$\langle u_2, v_1 \rangle = 0 + 0 + 0 + 2 + 0 = 2$$

$$v_2 = u_2 - \frac{\langle u_2, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \\ 0 \end{bmatrix} - \frac{2}{6} \begin{bmatrix} 1 \\ 0 \\ 0 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} -1/3 \\ 1 \\ 0 \\ -1/3 \\ -1/3 \end{bmatrix}$$

$$v_3 = u_3 - \frac{\langle u_3, v_2 \rangle}{\langle v_2, v_2 \rangle} v_2 - \frac{\langle u_3, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1$$

$$\langle v_2, v_2 \rangle = \frac{1}{9} + 1 + 0 + \frac{1}{9} + \frac{1}{9} = \frac{4}{3}$$

$$\langle u_3, v_2 \rangle = 0 + 0 + 0 + \frac{1}{3} + \frac{2}{3} = \frac{1}{3}$$

$$\langle u_3, v_1 \rangle = 0 + 0 + 0 + 2 - 2 = -4$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \\ -2 \end{bmatrix} - \frac{\frac{1/3}{4/3}}{\frac{4/3}{3}} \begin{bmatrix} -1/3 \\ 1 \\ 0 \\ -1/3 \\ -1/3 \end{bmatrix} - \frac{\frac{-4}{6}}{6} \begin{bmatrix} 1 \\ 0 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 + \frac{1}{12} + \frac{2}{3} \\ 0 - \frac{1}{4} + 0 \\ 1 + 0 + 0 \\ 1 + \frac{1}{12} + \frac{4}{3} \\ -2 + \frac{1}{12} + \frac{2}{3} \end{bmatrix} = \begin{bmatrix} 3/4 \\ -1/4 \\ 1 \\ -1/4 \\ -5/4 \end{bmatrix}$$

$$\text{Orthogonal Baz: } \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1/3 \\ 1 \\ 0 \\ -1/3 \\ -1/3 \end{bmatrix}, \begin{bmatrix} 3/4 \\ -1/4 \\ 1 \\ -1/4 \\ -5/4 \end{bmatrix} \right\}$$

$$\|v_1\| = \sqrt{6}$$

$$\|v_2\| = \frac{2}{\sqrt{3}}$$

$$\|v_3\| = \frac{\sqrt{52}}{4}$$

c) Orthonormal Baz: $\left\{ \begin{bmatrix} 1/\sqrt{6} \\ 0 \\ 0 \\ -2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix}, \begin{bmatrix} -\sqrt{3}/6 \\ \sqrt{3}/2 \\ 0 \\ -\sqrt{3}/6 \\ -\sqrt{3}/6 \end{bmatrix}, \begin{bmatrix} 3/\sqrt{52} \\ -1/\sqrt{52} \\ 4/\sqrt{52} \\ -1/\sqrt{52} \\ -5/\sqrt{52} \end{bmatrix} \right\}$

$\mathbb{R}^2$ deti standart baz: $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$, $L\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = x$, $L\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = y$ olun.

(3)

$$a) L\left(\begin{bmatrix} 3 \\ 2 \end{bmatrix}\right) = L\left(3 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = 3L\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) + 2L\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = 3x + 2y = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$L\left(\begin{bmatrix} 4 \\ 3 \end{bmatrix}\right) = 4x + 3y = \begin{bmatrix} 0 \\ -5 \\ 1 \end{bmatrix} \Rightarrow x + y = \begin{bmatrix} -1 \\ -7 \\ -2 \end{bmatrix} \Rightarrow 2x + y = \begin{bmatrix} -2 \\ -14 \\ -4 \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & -4 \\ 16 & -23 \\ 7 & -9 \end{bmatrix} \Rightarrow L\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 3 & -4 \\ 16 & -23 \\ 7 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad x = \begin{bmatrix} 3 \\ 16 \\ 7 \end{bmatrix} \Rightarrow y = \begin{bmatrix} -4 \\ -23 \\ -9 \end{bmatrix}$$

b) $\text{Ker } L = ?$ $L\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$. derden sistem;

$$\left[\begin{array}{cc|c} 3 & -4 & 0 \\ 16 & -23 & 0 \\ 7 & -9 & 0 \end{array} \right] \sim \left[\begin{array}{cc|c} 3 & -4 & 0 \\ 16 & -23 & 0 \\ 1 & -1 & 0 \end{array} \right] \sim \left[\begin{array}{cc|c} 0 & -1 & 0 \\ 0 & -7 & 0 \\ 1 & -1 & 0 \end{array} \right] \sim \left[\begin{array}{cc|c} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right] \quad \text{rank} = n$$

trivial solution

Dolayısıyla $\text{Ker } L = \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\}$. $\dim \text{Ker } L = 0$

c) $\dim L(\mathbb{R}^2) = ?$ $\dim \text{Ker } L + \dim L(\mathbb{R}^2) = \dim \mathbb{R}^2$
 $0 + \dim L(\mathbb{R}^2) = 2 \Rightarrow \dim L(\mathbb{R}^2) = 2$

ispat: $\begin{bmatrix} 3 & -4 \\ 16 & -23 \\ 7 & -9 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \rightarrow \text{Bazis} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\} \Rightarrow \dim L(\mathbb{R}^2) = 2 //$

d) $\langle A, B \rangle = a_1 b_1 + 2a_2 b_2 + 3a_3 b_3 + 4a_4 b_4$

i) $\langle A, A \rangle = 0 \Rightarrow A = 0$

$\langle A, A \rangle = a_1^2 + 2a_2^2 + 3a_3^2 + 4a_4^2 = 0 \Rightarrow a_1 = a_2 = a_3 = a_4 = 0 \Rightarrow A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

ii) $\langle A, B \rangle = \langle B, A \rangle$

$\langle A, B \rangle = a_1 b_1 + 2a_2 b_2 + 3a_3 b_3 + 4a_4 b_4 = b_1 a_1 + 2b_2 a_2 + 3b_3 a_3 + 4b_4 a_4 = \langle B, A \rangle$

iii) $\langle A+B, C \rangle = \langle A, C \rangle + \langle B, C \rangle$

$\langle A+B, C \rangle = (a_1 + b_1)c_1 + 2(a_2 + b_2)c_2 + 3(a_3 + b_3)c_3 + 4(a_4 + b_4)c_4$
 $= (a_1 c_1 + 2a_2 c_2 + 3a_3 c_3 + 4a_4 c_4) + (b_1 c_1 + 2b_2 c_2 + 3b_3 c_3 + 4b_4 c_4)$
 $= \langle A, C \rangle + \langle B, C \rangle$

iv) $\langle \alpha A, B \rangle = \langle A, \alpha B \rangle = \alpha \langle A, B \rangle$

$= \alpha (a_1 b_1 + 2a_2 b_2 + 3a_3 b_3 + 4a_4 b_4) = \alpha \langle A, B \rangle$

$\langle \alpha A, B \rangle = (\alpha a_1)b_1 + 2(\alpha a_2)b_2 + 3(\alpha a_3)b_3 + 4(\alpha a_4)b_4 = \alpha (a_1 b_1 + 2a_2 b_2 + 3a_3 b_3 + 4a_4 b_4) = \alpha \langle A, B \rangle$